



DEFENDER ARTHA WIDIPRASETYO

COMPUTER ENGINEERING GRADUATE

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Computer Engineering graduate from Institut Teknologi Sepuluh Nopember (ITS) with hands-on experience in software development, web applications, IoT systems, and Artificial Intelligence through internships, academic projects, and freelance work. Experienced in developing interactive and scalable applications, designing system architectures, and implementing AI-based solutions, including a multimodal question answering system using Qwen and Retrieval-Augmented Generation (RAG). Strong in analytical problem-solving, system development, and technical communication, with additional experience in content writing and copywriting.

EDUCATION

Institut Teknologi Sepuluh Nopember

2022 - 2026

- Bachelor's Degree in Computer Engineering
- GPA: 3.40/4.00
- Final Project: Multimodal Question Answering on Images and Audio Based on Qwen Model Using Retrieval-Augmented Generation (RAG)

SMA N 1 Salatiga

2019 - 2022

- Science Major (Natural Sciences & Mathematics)
- Ranked 31st out of 300+ students in the final overall ranking
- Participated in the National Science Olympiad (Informatics) for 2 consecutive years

WORK EXPERIENCE

Production Engineer | PT Bukit Makmur Mandiri Utama

Feb 2026 – Jul 2026

Internship

- Analyzed production performance metrics (utilization, payload) to support operational decision-making across mining activities
- Supported root cause analysis to identify performance gaps and contributed to improvement initiatives in mining operations
- Monitored production data and evaluated operational plans to help improve efficiency and target achievement
- Developed a web-based **bench height monitoring system** to automatically measure and report mining bench height, improving accuracy and efficiency in operational monitoring
- Contributed as a **technology implementer** by integrating digital solutions into operational mining workflows to support data-driven decision-making

IoT Engineer | PT Javadwipa Duta Mandiri

Sep 2025 – Dec 2025

Internship

- Developed an IoT-based energy monitoring system that collects and transmits real-time electricity consumption data to a centralized web dashboard.
- Focused on the IoT hardware development, including sensor integration, microcontroller programming, and data communication using efficient network protocols.

Front-end Developer | PT Teknologi Kartu Indonesia

Jul 2025 – Aug 2025

Internship

- Revamped the company's Learning Management System (LMS) interface based on new UI/UX designs to improve user accessibility and engagement.
- Initiated additional intuitive visual elements and interactive effects beyond the original design through collaboration with the product manager, **enhancing the overall user experience**.

Laravel Web Developer | PT Winnicode Garuda Teknologi

Mar 2025 – Aug 2025

Internship

- Designed and developed a full-stack job portal using Laravel, starting from UI/UX planning in Figma to full implementation.

Wordpress Web Developer | PT Damases Sejahtera

Mar 2025 – July 2025

Internship

- Redesigned key page layouts with modern design principles, **resulting in a fresher and more professional brand presentation.**
- Ensured consistency in typography, color scheme, and responsive layout across various devices and screen sizes.

Copywriter | Tim Media Teknik Komputer ITS

Feb 2025 – Feb 2026

Contract

- Responsible for all copywriting needs within the Computer Engineering department, including event news articles, social media captions, announcement broadcasts, and brochures.
- Crafted compelling and engaging content to enhance departmental communication and public outreach, leading to a **significant improvement in the overall quality of written materials** across all Computer Engineering media channels.

Full Stack Developer | Dinas Komunikasi dan Informatika Kab. Semarang

Jan 2025 – Feb 2025

Internship

- Collaborated with a team member to design and develop the "Komdigi Executive Dashboard" web application within one month.
- Engineered key system architecture components, including the Entity-Relationship Diagram (ERD) and Activity Diagram, to define workflows and database structure.
- Designed an intuitive UI/UX, with positive feedback from the Head of the Application Informatics Division, noting its **smooth and user-friendly experience.**
- Developed and optimized the front-end using modern frameworks and tools, receiving praise for its **fast performance** due to server-side rendering (SSR).
- Authored technical documentation for both users and developers to facilitate future maintenance and scalability.

Content Writer | Mojok.co

Aug 2024 – Apr 2025

Freelance

- Wrote engaging automotive articles, covering vehicle reviews and consumer insights.
- One of my articles ranked in the **Top 5 most-read weekly articles**, later featured in Mojok.co's Reels, **gaining 260,000+ views.**
- Applied SEO best practices, optimizing content for higher search rankings and audience engagement.

Content Writer | IDN Times

May 2019 – Mar 2023

Freelance

- Wrote and **published 125 articles** on IDN Times, covering product reviews, buying guides, and industry updates in gadgets and automotive.
- Conducted in-depth technical research to provide accurate and engaging content for a wide audience.
- Achieved a total of **538,000+ views** and **ranked 487th out of 98,593 writers** on IDN Times.

ORGANIZATIONAL EXPERIENCE

Head of Student Welfare | HIMATEKKOM ITS

Mar 2025 – Feb 2026

- Oversee and coordinate all student welfare programs, focusing on academic support, mental well-being, and inclusive campus engagement.
- Lead a team of staff members to execute key initiatives.

Staff of Student Welfare | HIMATEKKOM ITS

Mar 2024 – Feb 2025

- Supported student well-being by addressing academic, mental health, and social needs through various initiatives.
- Managed the "Bank Soal Teknik Komputer" program, successfully **achieving 100% KPI** and providing essential academic resources to support students' learning needs.

Member | Technopreneurship Development Center (TDC) ITS**Sep 2023 – Sep 2024**

- Participated in business training and entrepreneurship webinars, gaining insights into startup development, market strategies, and technological innovation in business.

HONORS AND AWARDS

2nd Place KSE Business Plan Competition by Diponegoro University	Mar 2024
Top 5 Finalist DIG IN Business-IT Case Competition by Binus University	Feb 2024
Finalist and Best Poster RISE Business Case Competition by ITS	Oct 2024
Silver Medalist Diponegoro University Science Competition	Oct 2024
3rd Place National Science Olympiad in Informatics (City Level)	May 2020

For a complete list of my projects, visit: builtbydefender.vercel.app